

MILESTONE 100% COMPLETED

IntellMeet — Backend Workstream Milestone Report

Member 2 (Backend, Database & Real-Time) • Week 1 Execution Plan Fully Completed & Verified

PROJECT

IntellMeet (MERN + AI)

TEAM MEMBER

Member 2 (Backend Lead)

MILESTONE STATUS

Week 1: 100% Complete

AUDIT TIMESTAMP

September 2026 (Live Audit)

WEEK 1 COMPLETION

100%

All Days 1 to 7 objectives met

AHEAD-OF-SCHEDULE

4 Modules

WebRTC, Live Chat, Tasks & Admin

PENDING GAPS

0 Left

All identified gaps closed

BUILD STATUS

Passing

TypeScript 0 errors, Git initialized

DAY-BY-DAY MILESTONE EXECUTION STATUS

DAY	PLANNED MILESTONE (28-DAY PLAN)	ACTUAL IMPLEMENTED CODEBASE STATUS	STATUS
Day 1	MERN boilerplate with Vite & Express; MongoDB setup; dependencies; folder structure; initial commit.	Express + TS server in <code>server.ts</code> . Mongoose connection with Memory fallback in <code>db.ts</code> . Git initialized with 107 files committed.	✓ 100% Completed
Day 2	User model & auth routes; JWT implementation with refresh tokens; password hashing with bcrypt.	<code>User.ts</code> with bcrypt pre-save hashing & roles. Register, Login, Refresh token rotation (<code>/api/auth/refresh</code>), and Logout.	✓ 100% Completed
Day 3	Profile creation & avatar upload via Cloudinary; protected routes with middleware; auth rate limiting.	Cloudinary SDK with Multer stream & Data URI fallback in <code>config/cloudinary.ts</code> . Strict 10-req limiter on auth in <code>rateLimiter.ts</code> .	✓ 100% Completed
Day 4	Meeting model and basic CRUD for meetings; WebRTC setup for video calls (peer connection logic).	Full Meeting CRUD (Create, Read, Room join, Update, and <code>DELETE /api/meetings/:id</code>). Full WebRTC signaling in <code>socketHandler.ts</code> .	✓ 100% Completed
Day 5	Redis setup for session and meeting caching; Socket.io server configuration for real-time features.	Redis client in <code>config/redis.ts</code> with <code>CacheService.ts</code> for active room participants with resilient in-memory fallback.	✓ 100% Completed
Day 6	Basic chat functionality in meetings; Real-time notification setup using Socket.io events.	<code>Message.ts</code> chat persistence with typing indicators. Real-time push notifications over <code>emitRealtimeNotification</code> on task assignments.	✓ 100% Completed
Day 7	Week 1 Checkpoint: Backend running locally, auth working, meeting creation, Postman test, README.	Server compiles with 0 errors. Seed data automated (<code>seedData.ts</code>). Complete <code>IntellMeet_Postman_Collection.json</code> created.	✓ 100% Completed

🏆 Newly Completed Features (Previously Pending Gaps)

- ✓ **JWT Refresh Token Rotation:** Implemented `POST /api/auth/refresh` and revocation logic on user document in `authController.ts`.

🚀 Advanced Deliveries (Ahead of Schedule)

- ★ **WebRTC Signalling (Week 2):** Complete peer connection negotiation (`webrtc:offer`, `answer`, `ice-candidate`) ready for frontend video room.
- ★ **In-Meeting Collaboration Chat (Week 2):** Real-time messages, typing indicators, and room activity events

- ✓ **Cloudinary Profile Avatar Upload:** Integrated `cloudinary` + `multer` in `UserController.ts` (`POST /api/users/avatar`) with seamless offline fallback.
- ✓ **Meeting Delete Endpoint:** Implemented `DELETE /api/meetings/:id` with host/admin permission checks and `AuditLog` tracking.
- ✓ **Redis Caching Layer:** Created `redis.ts` & `CacheService.ts` to cache meeting participants and sessions with zero-downtime memory adapter.
- ✓ **Brute-Force Rate Limiting:** Added 10-attempt limiter on `/register` , `/login` , and `/refresh` in `rateLimiter.ts` .

saved to MongoDB and broadcast via `Socket.io`.

- ★ **Task & Kanban Board APIs (Week 3):** Full Task management CRUD with priority, due date, status workflow, and automatic notification dispatch.
- ★ **Enterprise Audit & Governance:** Central audit logging system tracking logins, registrations, and meeting operations.

ARCHITECTURE SUMMARY & VERIFICATION MATRIX

MODULE / FEATURE	IMPLEMENTATION FILE	API ENDPOINT / MECHANISM	TEST STATUS
Authentication & Refresh	controllers/authController.ts	POST /api/auth/register POST /api/auth/login POST /api/auth/refresh	Verified ✓
Profile & Avatar Upload	controllers/userController.ts config/cloudinary.ts	POST /api/users/avatar (Multipart/form-data via Cloudinary SDK)	Verified ✓
Meeting Management (CRUD)	controllers/meetingController.ts	GET, POST /api/meetings GET, PUT, DELETE /api/meetings/:id	Verified ✓
Redis Cache / State	config/redis.ts services/cacheService.ts	CacheService.addRoomParticipant() CacheService.getRoomParticipants()	Verified ✓
Real-Time Signaling & Chat	sockets/socketHandler.ts	webrtc:offer, webrtc:answer, chat:send, user:join	Verified ✓
Push Notifications	controllers/notificationController.ts	emitRealtimeNotification() to user:userId room	Verified ✓
Brute-Force Rate Limiting	middleware/rateLimiter.ts	10 req/15min on auth endpoints, 1000 req/15min on general API	Verified ✓
Postman API Collection	IntellMeet_Postman_Collection.json	Complete JSON collection with preconfigured requests & variables	Verified ✓

🚀 Ready for Member 1 & Member 3 Week 2 Handoff

With all Week 1 deliverables 100% completed and key Week 2 real-time sockets already in place:

- **Member 1 (Frontend):** Can directly connect the Meeting Lobby, WebRTC video tiles, chat drawer, and Cloudinary avatar uploads to the working backend endpoints.
- **Member 3 (AI/DevOps):** Can integrate OpenAI transcription streams directly into meeting session endpoints with full confidence in authentication and state persistence.
- **Repository Status:** Local Git repository initialized with all 107 files committed (`c4055a6`).